





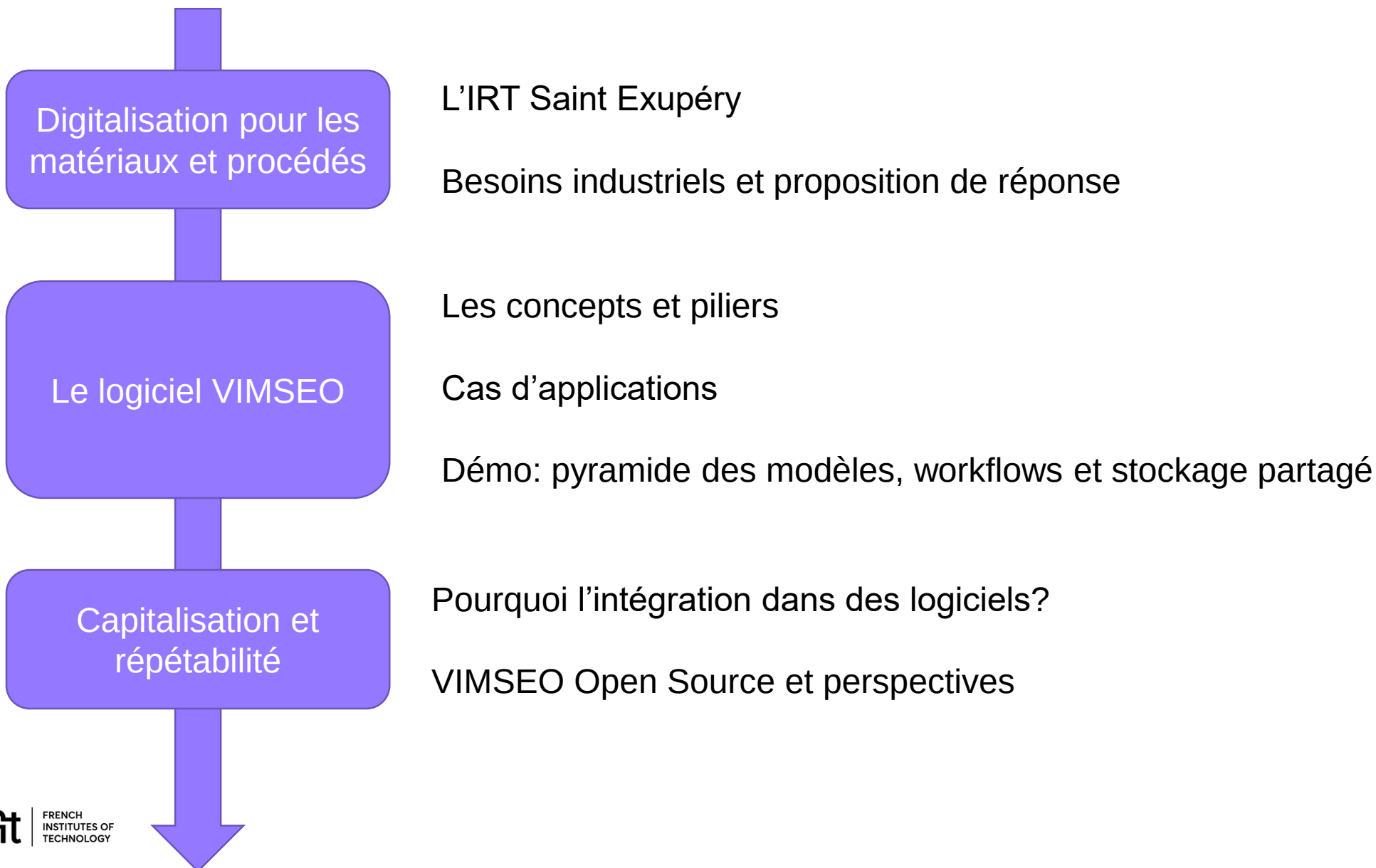
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Digitalisation pour les matériaux et les procédés : logiciel VIMSEO

Bénédicte REINE, Sébastien BOCQUET

GT Capitalisation et partage des connaissances matériaux, MECAMAT, 23/06/2026

Déroulé



Key figures (2025).



RESEARCH @IRT

39 M€ Annual budget

63 On going projects including

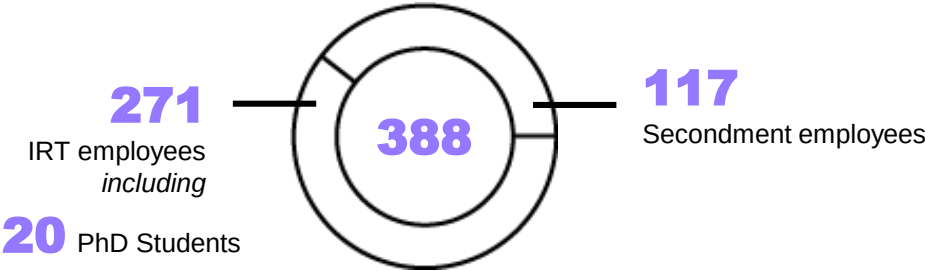
12 European projects

3 Start-ups

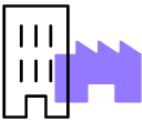
171 Publications & Communications in 2024

11 Platforms

PEOPLE



MEMBERS & PARTNERS



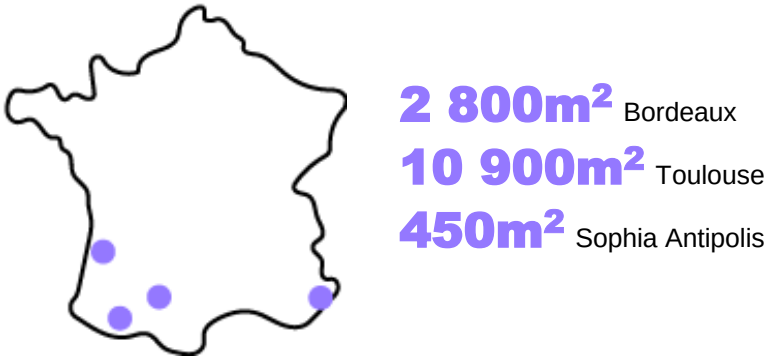
126 Industrial Members



30 Academic Members (supervisory bodies and public institutions) representing

43 Partner Laboratories

SITES



Marchés et synergies.

Marchés cibles



Aéronautique



Espace

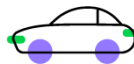


Defense

Marchés « synergies »



Ferroviaire



Mobilité terrestre



Énergie



Médical



Environnement / météo



Maritime

Notre stratégie

Technologies plus vertes

TPV



- > Énergie haute tension
- > Conversion de puissance électronique fiable, dense et efficiente

Technologies de fabrication avancées

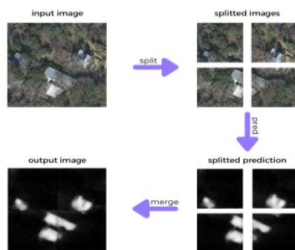
TFA



- > Matériaux métalliques et procédés
- > Surfaces et assemblages
- > Matériaux composites

Technologies intelligentes

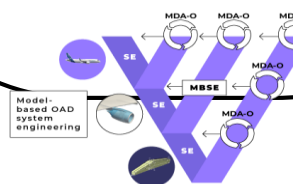
TI



- > Apprentissage avancé
- > IA pour les systèmes critiques
- > Connectivité & capteurs intelligents

Méthodes et outils pour le développement de systèmes complexes

MOSC



- > Ingénierie Des Systèmes
- > Optimisation Multi-Disciplinaires
- > Systèmes Embarqués Critiques



Digital approaches for increased maturity of new materials and processes

Context: New technologies enabling Cleaner Aviation



“Clean Aviation aims to develop, integrate and demonstrate disruptive technological innovations into new aircraft concepts by 2030-2035”.
+30% energy efficiency by 2035. Paving the way towards « Net-zero emission aviation by 2050 »

What does that mean for industry?

 **Introduce disruptive technologies (H2, Electric)**

Reduce development time 

 **Time to certification**

Maintain safety and reliability



New designs
 (e.g. H2 tanks, FoT, WoT)

New threats
 (e.g. H2, High voltage, Supply chain)

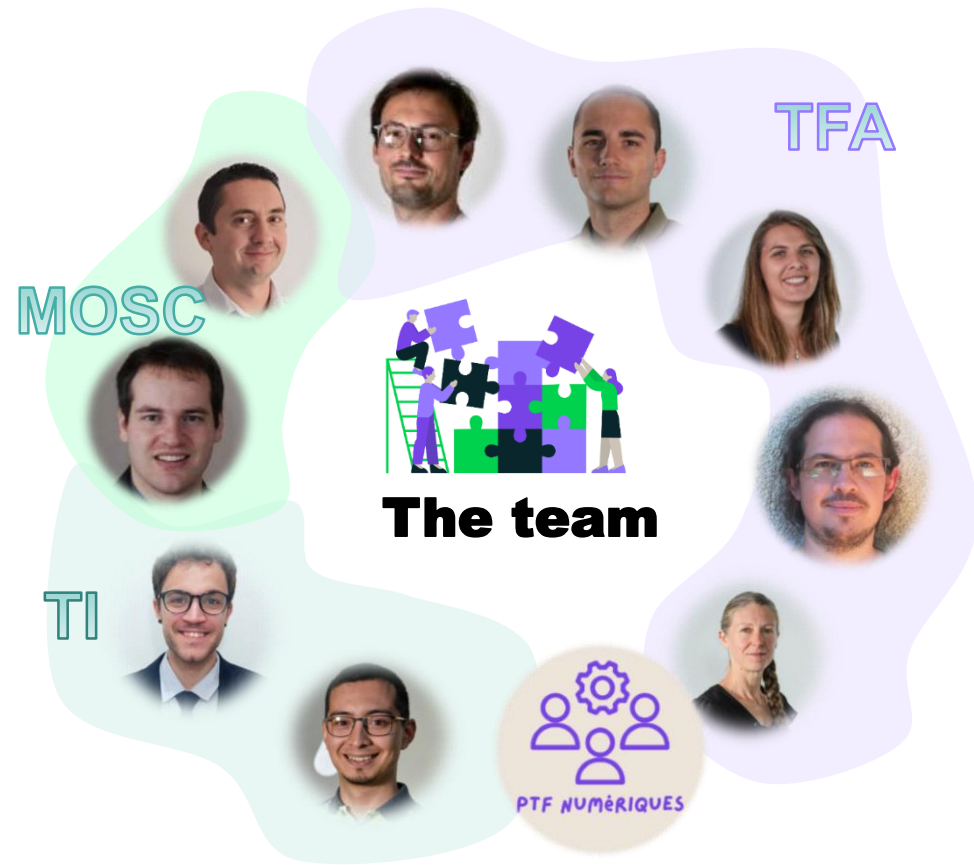


New manufacturing
 (e.g. OoA, Welding, AM, H2H)

New materials
 (e.g. bio-sourced, recycled)

How to reach TRL6 (and beyond) even faster?

Digitalisation as a key enabler

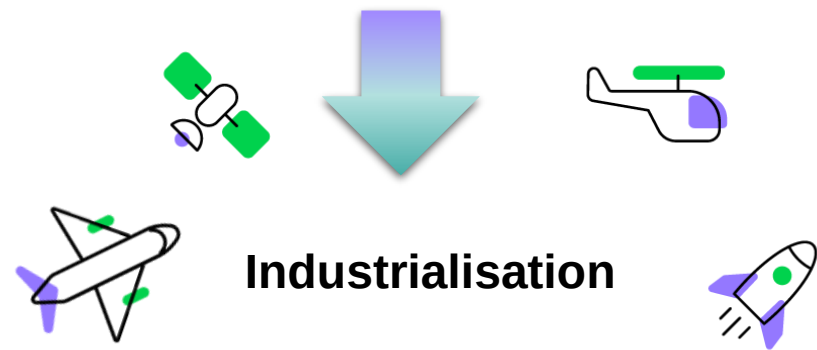


- ➡ Physics and Data driven modelling
- ➡ Application/development of analysis methods
- ➡ Decision-making support
- ➡ Integration and Interoperability (digital/digital & digital/physical)

+

Academic ecosystem & Partners

FRENCH INSTITUTES OF TECHNOLOGY



Model-based Decision Making

Qualification of Simulations & Certification by Analysis

Context

- ❑ Ambition to reduce design life cycle for the next SMR program
- ❑ Future Architecture trade-offs based on innovative designs will require more simulations and models
- ❑ Future designs will be multi-functional and multi-physics for which overall performance will be complex to assess by tests.

Needs

- ❑ Maximize the value of models all along design cycles by :
 - ❑ Improving Model reliability & traceability
 - ❑ Enabling Model Re-use
 - ❑ Mastering variability and uncertainty into simulations
- ❑ Establish and deploy optimal test plan to support model and simulations V&V

Key Scientific & technical lockers

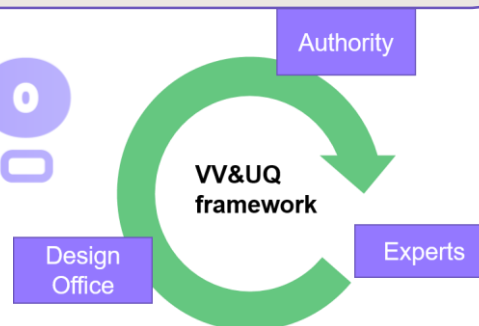
From unitary configurations to industrial mock-ups:

- ❑ Lack of consistency in model development
- ❑ Lack of numerical continuity
- ❑ Lack of continuous experimental database coverage
- ❑ Lack of confidence metrics
- ❑ Extensive costly testing without guarantee of reliable calibration

Our solution

Develop and embed into a unique and collaborative framework to foster model and simulation

VIMSEO



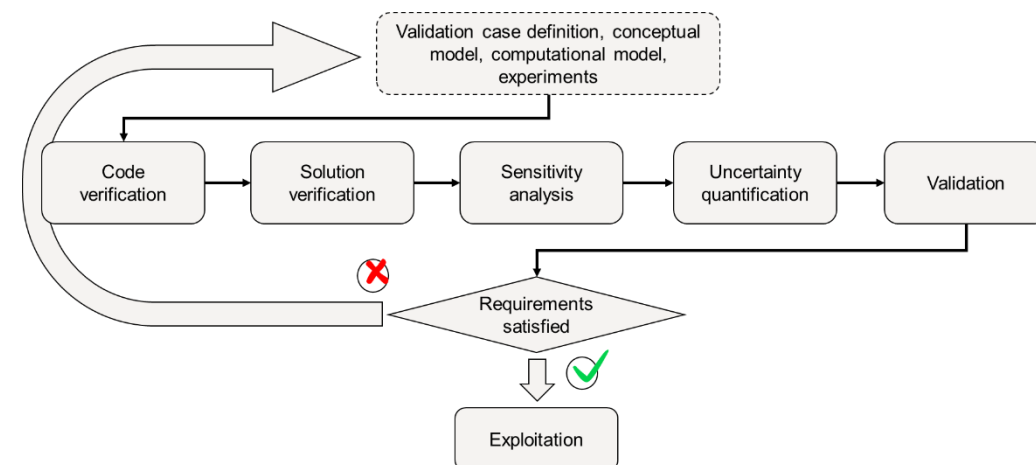
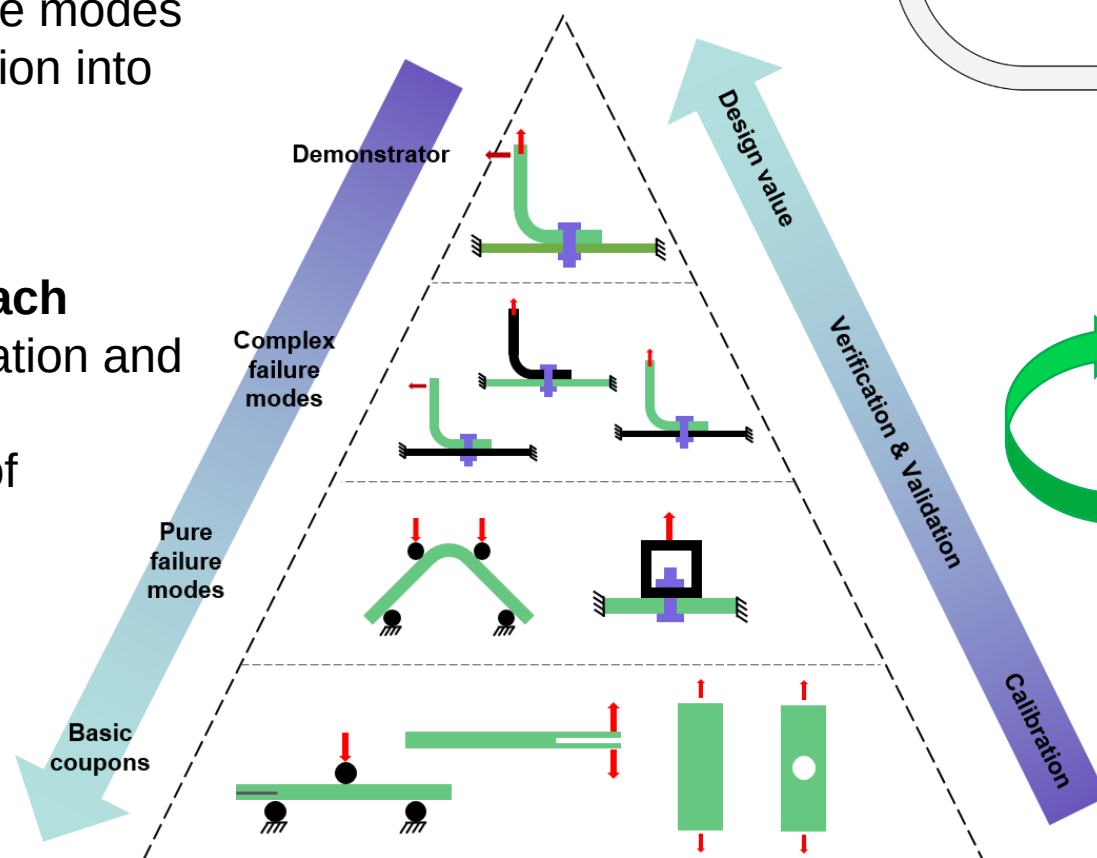


The VIMSEO software

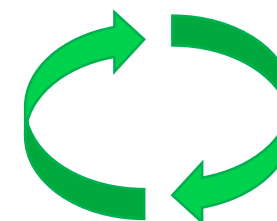
Three concepts in synergy

Building block approach

- **Top-down approach**
 - Complex failure modes
→ decomposition into unit tests for identification (calibration)
- **Bottom-up approach**
 - Models verification and validation
 - Computation of B-values



UQ&M, V&VUQ workflows



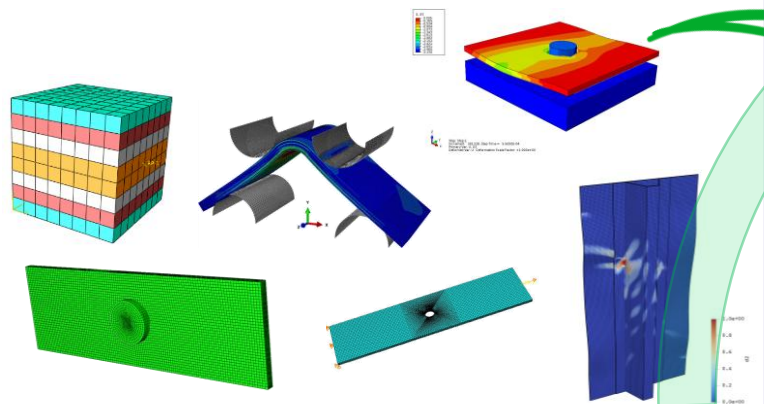
Model implementation
(numerical choices: type of EF, meshing, material law, solver etc...)

Load cases

VIMSEO Features

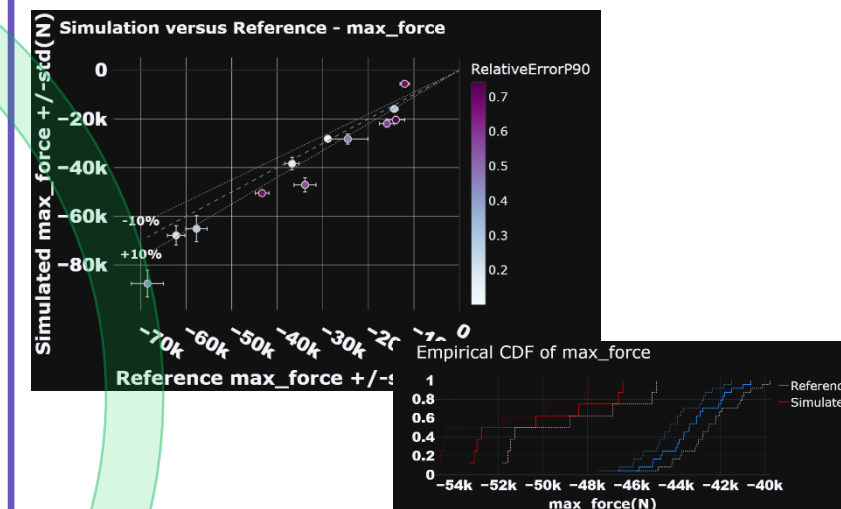
Pillar #1: a framework to integrate models

Capitalise parametric models



Pillar #2: a framework to define analysis

Capitalise analysis nodes



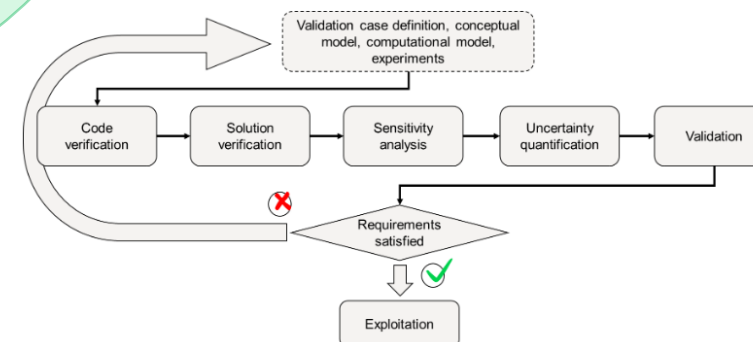
Pillar #4: platform-oriented services

Scale to execute VV&UQ workflows



Pillar #3: VV&UQ workflows + dashboards

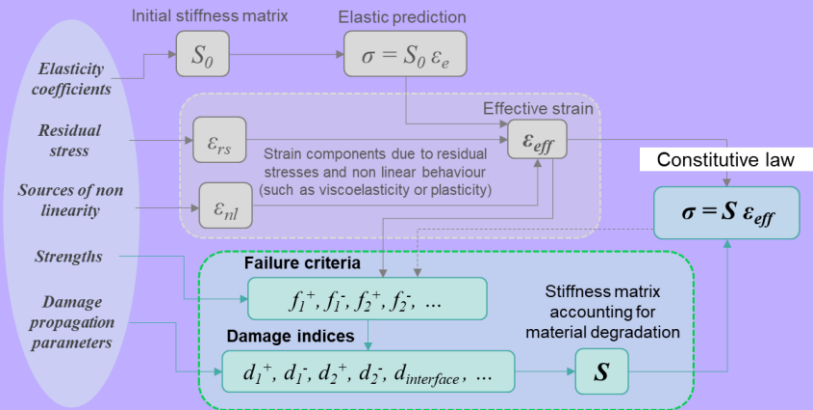
VV&UQ methodologies implemented as workflows



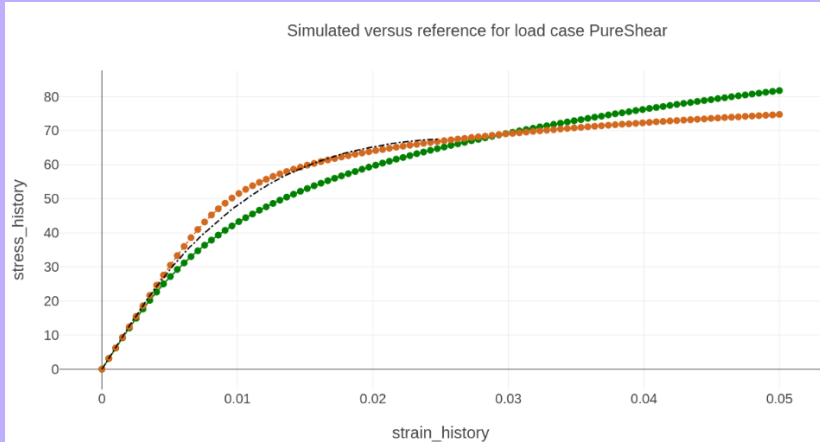
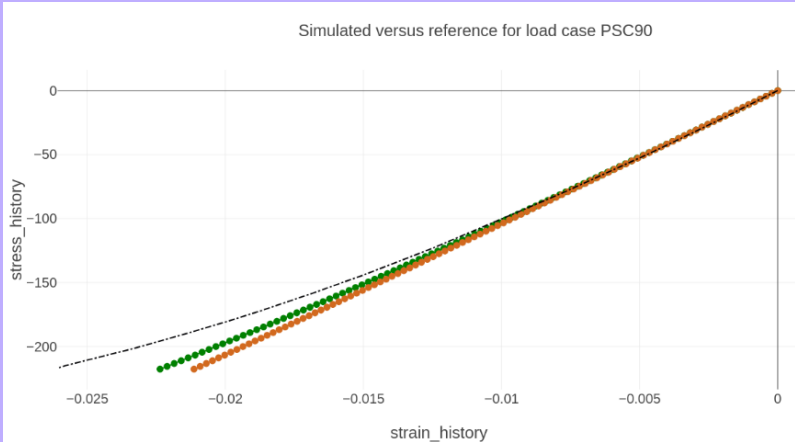
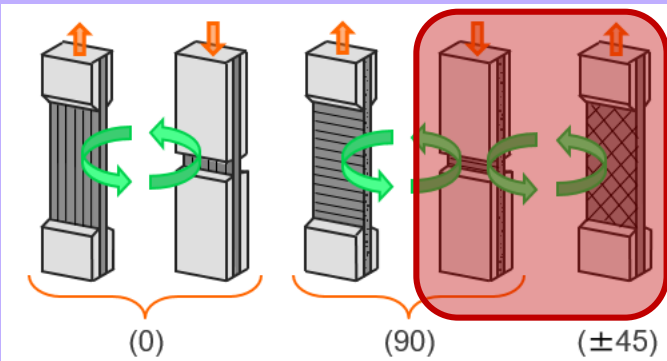
Use cases

I want to calibrate a model having parameters that are coupled across several load cases

=> *Multi-load case calibration with curve matching*



Standard	Description	Influent parameters
EN2561 type B	Longitudinal <u>tensile</u> strength	$E_1, E_{1t}, \varepsilon_{1t}, \nu_{12}, X_t$
EN2850 type B	Longitudinal <u>compressive</u> strength	$E_1, E_{1c}, \varepsilon_{1c}, X_c$
EN2597 type B	Transverse tensile strength	E_2, Y_t, n_0
EN2850 type B	Transverse compressive strength	$Y_c, E_2, n_0, \beta_{22}, \gamma, p_v$
AITM 1.0002	In plane shear strength	$G_{12}, n_0, \beta_{66}, \gamma, p_v, SF_{12}$

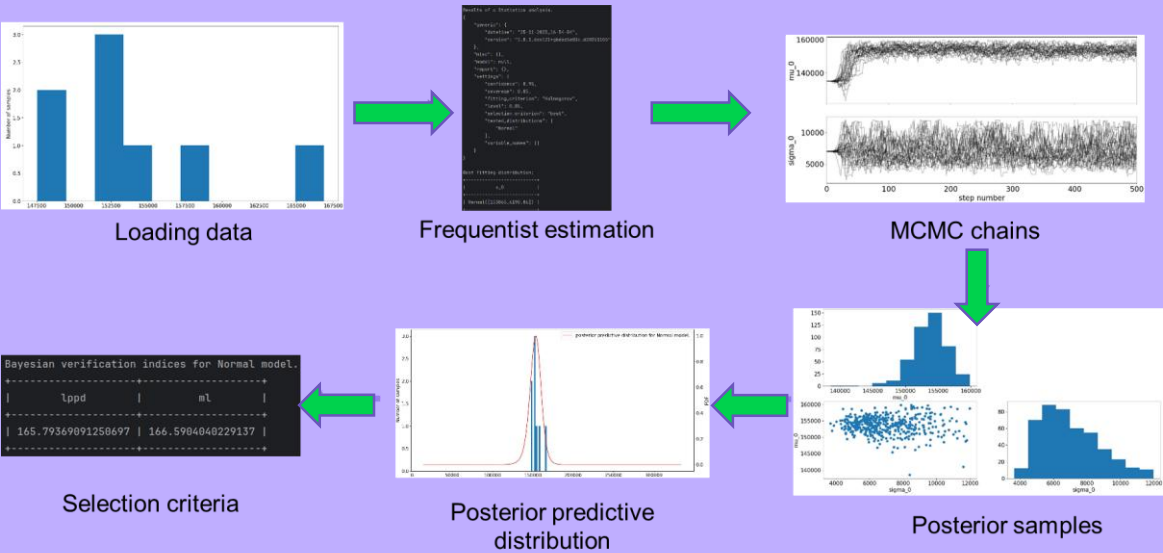


Onera Progressive Failure Model [Laurin, 2005]

Use cases

I want to quantify uncertainties on identified material properties

=> *Bayesian calibration*



I want to find the best sequence of layups for composites coupons in order to better characterize mechanical properties

=> *Optimization of identification tests sequences*

Stacking sequence	Laminate thickness
$[-15^\circ / +15^\circ / +15^\circ / -15^\circ]$ $[-30^\circ / +30^\circ / +30^\circ / -30^\circ]$ $[-45^\circ / +45^\circ / +45^\circ / -45^\circ]$ $[-60^\circ / +60^\circ / +60^\circ / -60^\circ]$ $[-75^\circ / +75^\circ / +75^\circ / -75^\circ]$	7.2 mm
$[-15^\circ / +15^\circ / 90^\circ_3 / +15^\circ / -15^\circ]$ $[-30^\circ / +30^\circ / 90^\circ_3 / +30^\circ / -30^\circ]$ $[-45^\circ / +45^\circ / 90^\circ_3 / +45^\circ / -45^\circ]$ $[-60^\circ / +60^\circ / 90^\circ_3 / +60^\circ / -60^\circ]$ $[-75^\circ / +75^\circ / 90^\circ_3 / +75^\circ / -75^\circ]$	12.6 mm

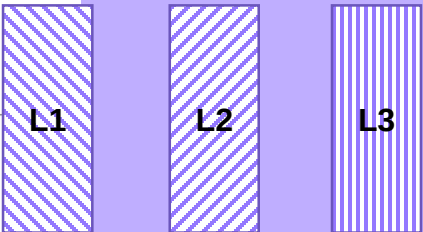


Table 4. Selected stacking sequences

Which lay-up?

What is the best sequence based on 3 lay-up?

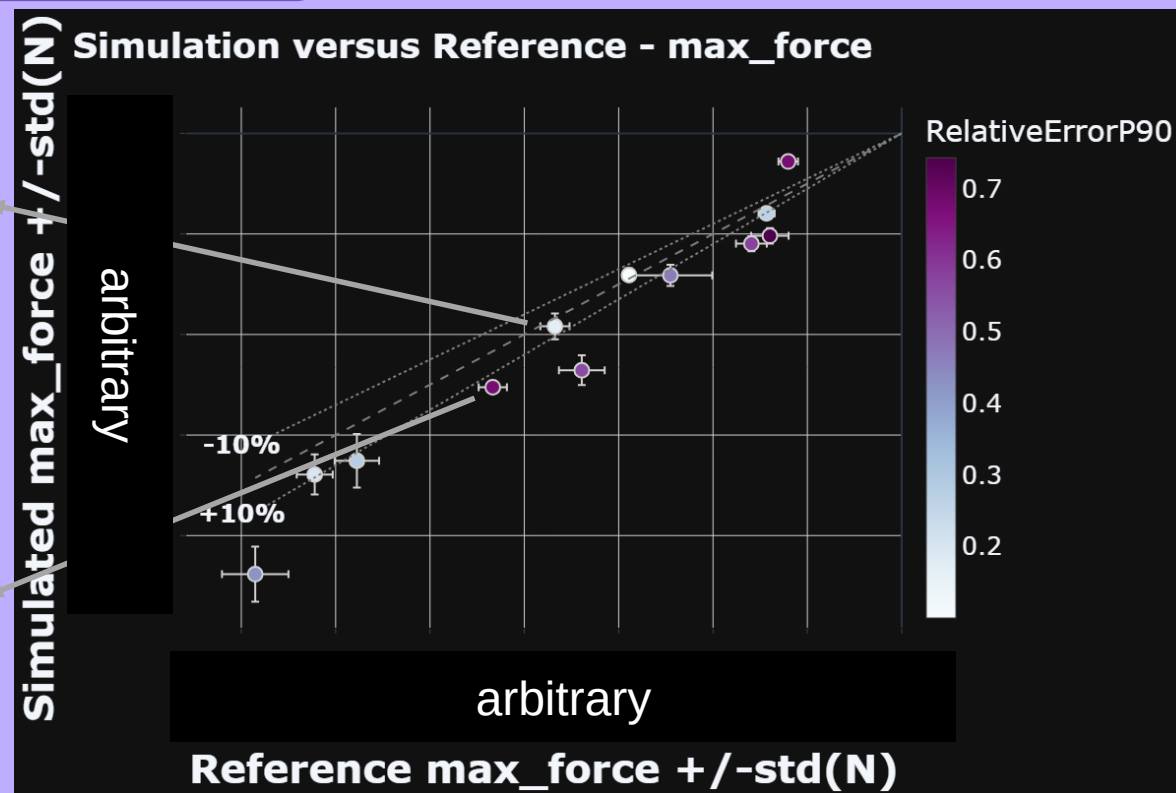
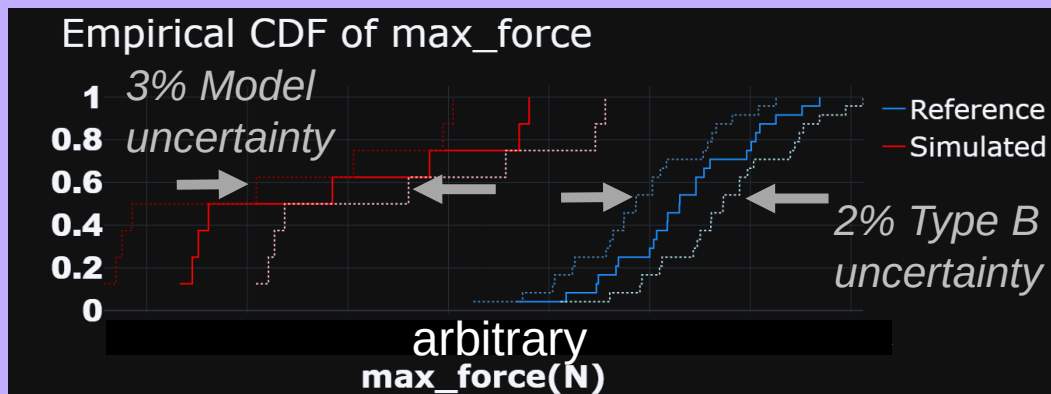
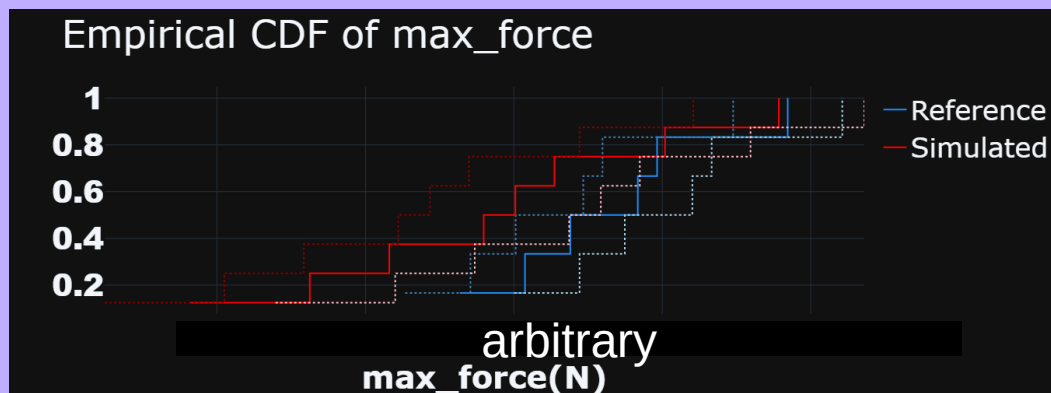
Use cases

I want to assess the risk of using a model, based on validation metrics containing both the bias and variability

=> *Stochastic validation*



Open Hole composites coupon in compression



Structuration autour des logiciels

References

EU project

Construction Of Novel CERTification methOds and means of compliance for disruptive technologies



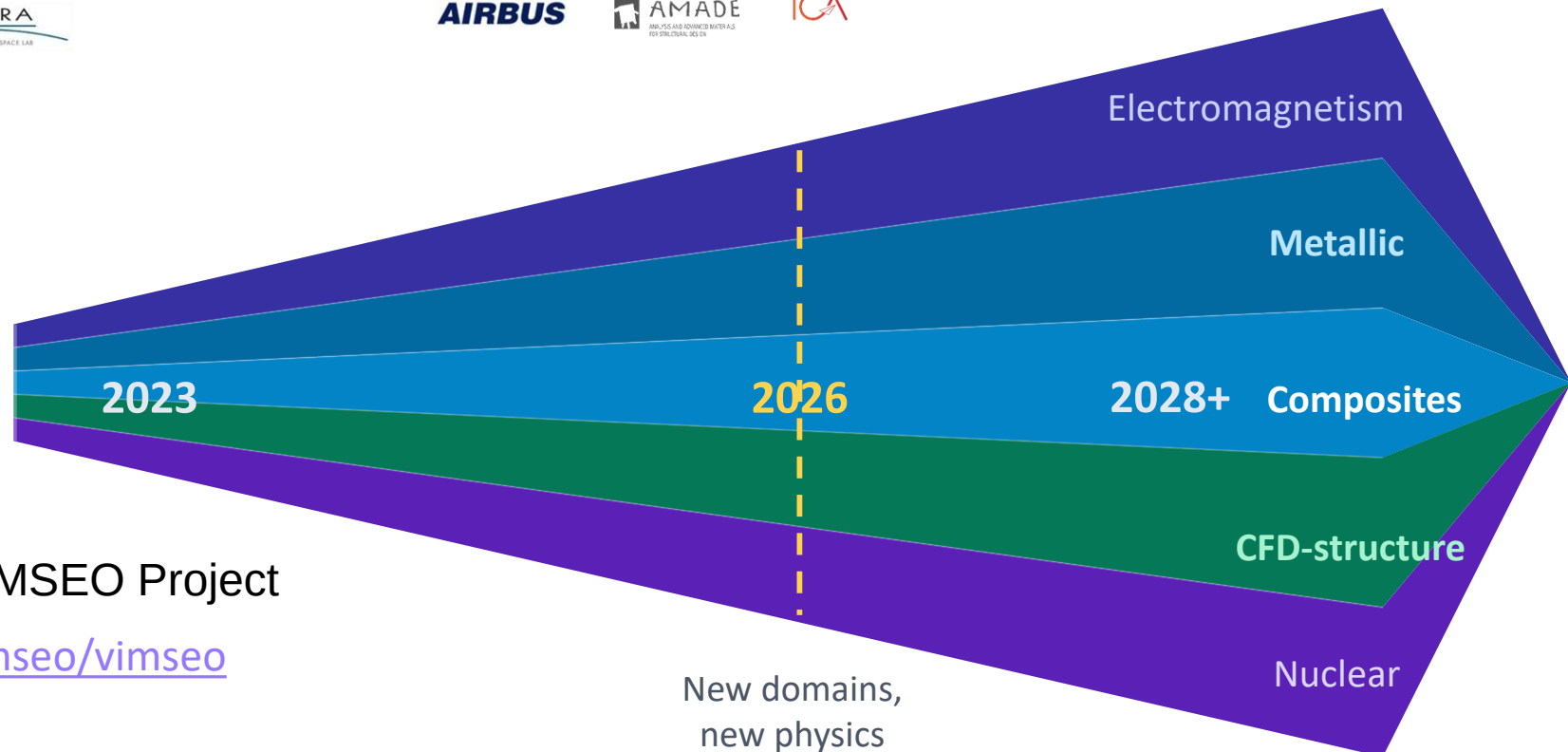
VITAL
2023-2026
Composites (damage, coupons, AFP)



TRUST
2022-2025
Composites (damage, coupons, AFP)



SUNRISE
2026-2029
Composites (damage, coupons with defects) CFD-structure sizing



The Open Source VIMSEO Project

<https://github.com/vimseo/vimseo>

[Overview - vimseo](#)

[Use a model - vimseo](#)



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DUCTILE
2026-2029
Aluminium (ductilité, fissuration)

